

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Operating Systems ■ CS-3009 ■ Autumn-2024, End Sem
■ 50 Marks ■ Time: 3hrs ■ 2 Pages ■ Answer ALL Questions
Ensure that all parts of a question are answered sequentially.

1. (a) What is multiprocessing in operating systems? How is it different from Multiprogramming? Draw and explain in brief. [2]
(b) Consider a system with N processors and P processes.
 - i. What is the maximum and minimum number of processes that can be in the Ready state?
 - ii. What is the maximum and minimum number of processes that can be in the Running state?Support your answers with proper reasoning. [2]
(c) What are the different methods for handling deadlock in OS, and which method is generally considered the most effective for ensuring system stability and performance and why? [2]
(d) Why a CPU scheduling policy suffers from starvation? Suggest a solution for the starvation problem with a suitable example. [2]
(e) Consider a system with a counting semaphore initialized to 7. If 4 processes perform a wait operation and 2 processes perform a signal operation, what will be the final value of the semaphore? [2]
2. (a) Explain the *Convoy Effect* with an example. A computer has *TEN* tape drives, with N processes competing for them. Each process may need *TWO* drives. What is the maximum value of N for the system to be deadlock-free? [5]
(b) Consider a system with FIVE processes [P1, P2, P3, P4, P5] with burst times [3, 4, 1, 5, 2] and arrival times [3, 4, 5, 6, 7]. If the system uses the Shortest Job First (SJF) scheduling algorithm then draw the Gantt chart and find the average waiting time of each process. Briefly explain if SJF scheduling policy suffers from starvation problem or not. [5]
3. (a) Explain the criteria for deadlock. What do you mean by a *safe sequence of execution*. Consider a system with 3 resource types A, B, and C. Four processes P_0, P_1, P_2 , and P_3 execute concurrently. The MAX and ALLOCATION matrix is shown as below. After current allocation, 3 instances of A and 3 instances of B are only resources available.

	MAX			ALLOCATION		
	A	B	C	A	B	C
P0	4	3	1	1	0	0
P1	2	1	4	1	0	2
P2	1	3	3	1	0	3
P3	5	4	1	2	0	0

Verify if the current state is *safe* or not. Show the steps clearly and corroborate your answer with proper explanation. [5]

- (b) Consider a system that follows 39-bit logical addressing. The page size is 4KB and page table entry size is 8 bytes. Give the reason why the system must implement hierarchical paging? How many levels of paging is required. Calculate the memory requirement of the page tables in each level. Show the fields of the logical address and mention the size of each field in bits. [5]
4. (a) Let the system follows dynamic partitioning scheme for contiguous allocation of memory to the processes. The processes are allocated memory for the required time-stamp of execution (Usage time), after which they terminate releasing the memory. After long interval of time, at time instance t , assume that there are 4 free-holes in memory of sizes 4KB, 8KB, 20KB, and 2KB. Free-holes are allocated using best-fit strategy. The allocation requests are stored in a queue as shown below.

Process No.	P1	P2	P3	P4	P5	P6	P7	P8
Process Size	2KB	14KB	3KB	6KB	6KB	10KB	7KB	20KB
Usage time	4s	10s	2s	8s	4s	1s	8s	6s

Show the snapshot of memory allocation and de-allocation for these requests on these free-holes, at different time instances. What is the time at which P7 will terminate? [5]

- (b) Consider a computer system with ten physical page frames. The system is provided with an access sequence $(a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20})$, where each a_i is a distinct page number. Show, how the frame allocations and replacements are done in Last In First Out (LIFO) page replacement strategy. Count the number of page faults in each case. Also, explain Belady's Anomaly, and explain with reasoning if it occurs in the LIFO page replacement strategy. [5]
5. (a) Consider there are five processes with burst times [14, 2, 5, 2, 2] arriving in a system that implements Multi Level Feedback Queue (MLFQ) scheduling. Their arrival times are 6, 8, 25, 27, and 29 respectively. There are three queues: Queue-1 (top level queue) implements RR scheduling with quanta 2. Queue-2 follows RR scheduling with quanta 5 and Queue-3 (bottom level queue) executes the processes in FCFS order. Depict the execution schedule of the processes and find the waiting time of each process. Assume context switch overhead time is zero. [5]
- (b) Differentiate between *global vs. local* page replacement. Explain why *thrashing* occurs in a system. How *Working set window* model helps in addressing thrashing issue? [5]